Uncanny Apology in Delivery Robot's Service Failure: How Mental Model and Interaction Design Impact Last-mile User Experience

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MOTIVATIONS

- **Research gaps in...**
  - Interaction between last-mile users and delivery robots
  - Service failure recovery strategies in delivery robots
- **Users' mental model of delivery robots**
  - How do users expect the delivery robot to appear and function?

STUDY DESIGN

- **Service failure recovery strategies in delivery robots**
  - Apology Strategy
  - Anthropomorphism (Facial Expressiveness)

APOLOGY STRATEGY

Simple Apology "Your food has arrived, I apologize for the delay."

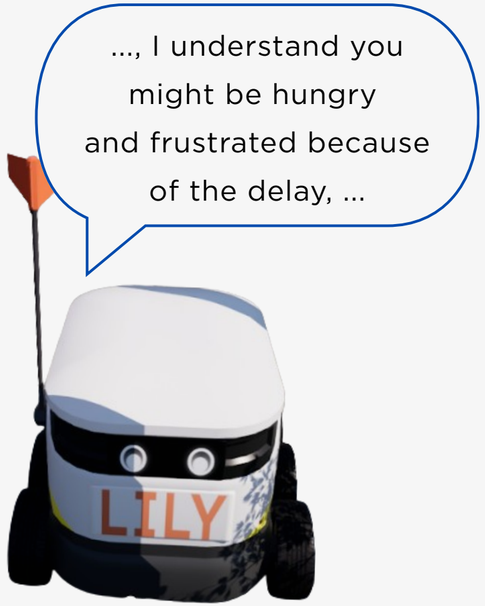
Empathic Apology "Your food has arrived, I understand you might be hungry and frustrated because of the delay, and I apologize for the delay."

Explanatory Apology "Your order has arrived. Due to an unexpected car accident, I had to reroute. I apologize for the delay."

FACIAL EXPRESSION LEVEL

High Expressiveness

Low Expressivenss

DISCUSSION

- **Effective recovery strategies…**
  - Explanatory apology
    - *Increasing functional trust*
  - High emotion expression level
    - *Amplify the authenticity of apologies*
- **Uncanny apology…**
  - Users experienced discomfort and less tendency to forgive when empathic apology given with low expressiveness